

CHAPTER 15

# Introduction to Risk Management

The principles and frameworks for identifying, measuring and controlling risk in oil trading.



## WHY IT MATTERS

# Oil is one of the most volatile exposures a business can carry

The price of oil drives the value of countless businesses — yet its volatility dwarfs almost every other market exposure they face. Left unmanaged, that price can single-handedly decide the fate of a company. Risk management is how they take back control.

**\$147 → \$51**

Brent collapsed within months in 2008 — a move that broke airlines and shippers alike

### **It reduces uncertainty**

Risk management smooths a company's cash flows — cutting both the downside and the upside — for a steadier business.

### **It is a choice, not a forecast**

It lets management dial exposure up or down to match their appetite — it is not a bet on where price goes next.

## THE IDEA

# What risk management is

It is the deliberate shaping of a company's risk profile — modifying and controlling the uncertainty in its cash flows so management can run the business on stable ground.

### Less uncertainty, both ways

Hedging trims the downside and the upside alike. You give up some windfall in exchange for protection — and predictability.

### A stable operating environment

Steadier cash flows let a company plan, invest and borrow with confidence — knock-on benefits that go far beyond the trade.

### Matched to risk appetite

Management chooses how much of each risk to keep — turning exposure into a deliberate decision rather than an accident.

### Not about making money

A hedge that loses money while the business gains is working perfectly. Risk management protects the core, it doesn't chase profit.

## KNOW YOUR ENEMY

# The relevant risks

Before managing risk you must name it. In oil, a handful of distinct exposures recur — each calling for a different response.

|                           |                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------|
| ■ <b>Commodity market</b> | Exposure to uncertain market prices, volatilities or correlations — the big one for oil.      |
| ■ <b>Proxy (basis)</b>    | The imperfect match between a hedge and the actual exposure being hedged.                     |
| ■ <b>Volumetric</b>       | Changes in the volume of commodity supplied or consumed — more or less than expected.         |
| ■ <b>Currency</b>         | Uncertainty in FX rates, since oil is priced in dollars and costs may be in other currencies. |
| ■ <b>Technical</b>        | A physical asset under-performing, leaving you unable to meet delivery or realise its value.  |
| ■ <b>Force majeure</b>    | A rare, unforeseen event with no contractual mitigation — the tail you cannot hedge away.     |

## THE PARTICIPANTS

# Who manages price risk

Every player along the oil chain carries a different exposure — and hedges to protect a different part of the business.

### End users

airlines · transport · utilities · shipping

- Fuel is 15–40% of operating costs
- Short physical products
- Need certainty for planning

### Refiners

processors

- Short crude, long products
- Exposed to the refining margin
- Margin drives whether to run

### Producers

upstream

- Fixed costs, 100% revenue exposure
- Long physical oil
- Need stability for capital projects

### Traders & exporters

merchants

- Multi-product exposure
- Margin protection & arbitrage
- Storage and time plays

## THE TOOLKIT

# What risk management can do

A risk programme is built from a menu of levers — combined to give the protection a company wants, at a cost it will accept.



### Lock in or cap

Fix a price outright, or cap the downside while keeping some upside — the core choice every hedge makes.



### Share the downside

Structures can pass part of the risk to a counterparty in exchange for giving up part of the gain.



### Build in flexibility

Volumetric flexibility, extendibility and cancellation rights let a hedge bend as the business changes.



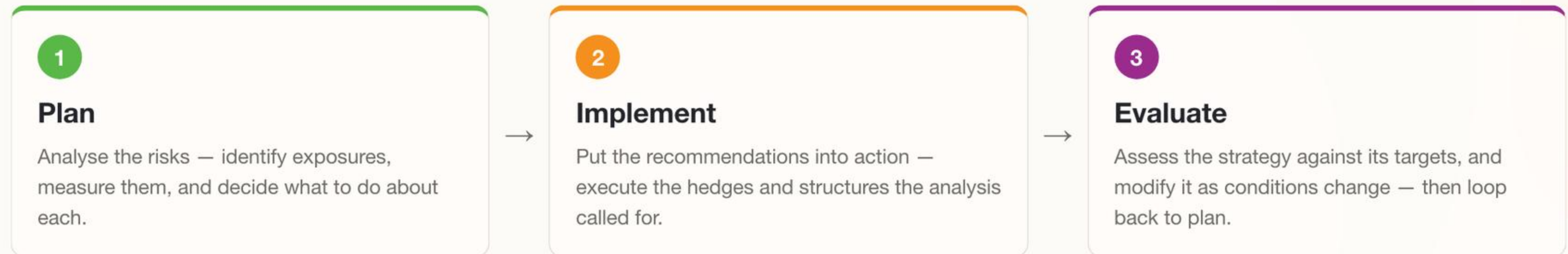
### Control timing & basis

Choose when decisions are made and switch between price indices to match the real exposure.

## AN ONGOING LOOP

# The risk management process

Risk management is not a one-off trade but a continuous cycle — three stages that repeat as the business and the market evolve.



↻ continuous — the programme is reviewed and adjusted, never "set and forget"

FROM PRINCIPLE TO PROGRAMME

# Designing a risk management programme

Turning the idea into a working programme means management has to put a clear structure in place — and govern it.

**1 Identify**

Map every risk the company is exposed to, and how they interact.

**2 Set policy**

Define the risk policy and aims, approved by management and investors.

**3 Quantify**

Measure the risks that can be measured to inform the decisions.

**4 Set limits**

Authority levels, risk limits, targets and budgets to be judged against.

**5 Implement & execute**

Build and run the programme with derivatives — within those limits.

**6 Monitor & modify**

Track performance, and adapt as the business and market evolve.

## TWO WAYS TO PLAY IT

# Tactical vs strategic

Every risk programme sits somewhere between two generic approaches — and most companies blend them.

### Tactical

**short-term · opportunistic**

React to current conditions — hedge when the price or the curve looks favourable, lift hedges when it doesn't. Nimble, but driven by the market of the moment.

### Strategic

**long-term · policy-driven**

Flow from the company's corporate strategy — a standing rule for how much to cover and when, designed to protect the business over years, not weeks.

## A STRATEGIC FRAMEWORK

# Strategic hedging — the value zone

A consumer hedges more aggressively when prices are attractive and the market is tight — and layers that cover by how far out the exposure sits.

### The value zone = a tight market

Low stocks, restricted supply, OPEC effective, supply & demand in balance — prices are supported, so a consumer locks cover in below budget.

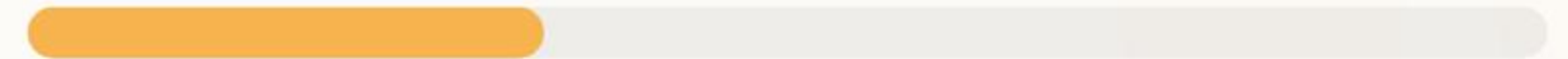
### Outside it, ease off

High stocks and abundant supply mean less urgency — maintain only short-term cover and keep flexibility.

### COVER BY HORIZON

#### 0–6 months

Maintain short cover



#### 6–18 months

Normal cover



#### 18+ months

Maximize cover



## HARD-WON LESSONS

# Pitfalls & misconceptions

When airlines slashed fuel hedges as oil fell from \$147 in 2008, the unhedged ones were savaged on the rebound — and Air China booked a ¥6.8bn paper loss hedging on a forecast. The recurring mistakes are always the same.

### "I hedged well — I made money on the paper trade"

A hedge isn't meant to make money; it offsets the physical. Judging it by its own P&L misses the entire point.

### "I should have hedged last week"

Risk management is a discipline, not market timing. Waiting for the perfect level is how exposures go unhedged.

### "The market will turn around!"

Hope is not a strategy. Refusing to act on a losing view is how a manageable risk becomes a crisis.

### Hedging on a forecast

Treating a price prediction as certainty — as some did at \$147 — turns a hedge into a leveraged bet.

CONCLUSION — CONTROL, NOT PREDICTION

# Risk management buys certainty, not profit.

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## **Name, measure, control**

Identify every exposure, quantify what you can, and shape the risk to fit your appetite.

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## **A process, not a trade**

Plan, implement, evaluate — governed by limits and reviewed as the business evolves.

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## **Discipline beats hope**

The cost of a hedge is the price of certainty; refusing to act is the most expensive choice of all.